SEID 2364 – Assignment 9 Mediation Pathways, Capability, and Responsibility

Part 1 – Narrative Description

An AI hiring system used by a large company automatically screened and ranked job applicants before recruiters reviewed applications. Applicants submitted resumes and personal information through an online platform. The AI system analyzed qualifications, work experience, and historical hiring patterns to determine candidate rankings.

Over time, concerns emerged that the system unfairly disadvantaged certain applicants, especially women and candidates from non-traditional educational backgrounds. Some qualified applicants were repeatedly filtered out before human review occurred. The harm affected applicants who lost employment opportunities and trust in the fairness of the recruitment process.

Part 2 – Originating Agents

1. Job Applicants 2. AI Hiring System 3. Company HR Department 4. Software Developers 5. Company Management

Part 3 – Mediation Pathways

Pathway 1 – Applicant Pathway

Applicant → Resume Submission → AI Screening → Recruiter Dashboard → Hiring Outcome

Space Types: - Personal: Applicant preparing resume - Remote: Online application platform - Virtual: AI processing system - Local: Recruiter reviewing recommendations

Pathway 2 – Developer Pathway

Developers → Algorithm Design → Training Data Integration → AI Model Deployment → Hiring Recommendations

Space Types: - Local: Development environment - Virtual: Machine learning systems - Remote: Cloud infrastructure - Local: Company HR systems

Part 4 – EDOCA Assessment

Transition 1 – Resume Data to AI Ranking

Effort: Applicants invest time and effort creating resumes, while the company provides computational resources for processing applications.

Observability: Applicants cannot fully observe how the AI transforms resumes into rankings. Recruiters have partial visibility, while developers have greater technical understanding.

Control: The company and developers control the ranking criteria and system updates. Applicants have almost no ability to modify outcomes once data is submitted.

Authority: Company management authorizes the use of the AI system and determines hiring policies.

Distortion: Important human qualities may become simplified into numerical scores, creating distortion between applicant abilities and AI evaluation.

Synthesis: The combination of low applicant control and low observability increases the risk of unfair outcomes and hidden discrimination.

Transition 2 – AI Recommendation to Hiring Decision

Effort: Recruiters review recommendations, but much of the evaluation work is delegated to automation.

Observability: Recruiters can observe rankings but may not fully understand how outputs were generated.

Control: Recruiters can technically override recommendations, but strong reliance on automation may reduce independent judgment.

Authority: HR departments and company leadership hold authority over final employment decisions.

Distortion: Bias within training data may distort recommendations and amplify existing inequalities.

Synthesis: When recruiters trust AI recommendations too strongly, responsibility becomes distributed and accountability becomes less clear.

Part 5 – Capability vs Intent

The company may claim that the system was designed to improve efficiency rather than discriminate. However, if harmful outcomes were foreseeable and safeguards were insufficient, the harm cannot be treated as completely unintended.

The AI system itself has no intent, but developers and organizations remain responsible for how the system is designed, deployed, and monitored.

Part 6 – Responsibility Distribution

Responsibility is distributed across multiple actors. Developers create the algorithms, companies deploy the system, recruiters apply recommendations, and management authorizes system use.

This distribution can create accountability gaps because different actors may shift blame onto each other when harmful outcomes occur.

Conclusion

This assignment showed that harm in AI hiring systems emerges through complex mediation pathways involving technical systems, institutions, and human actors.

Using EDOCA helped reveal how observability, control, authority, and distortion become unevenly distributed across the system. Ethical responsibility therefore cannot be reduced to a single actor because capability and decision-making are spread across the entire hiring structure.